

Word problems on ratio

Sharing an amount, scaling a recipe, and finding a quantity from a difference.

LESSON 44–46

3 × 40 MIN

MATEMATIKA 6, §10

S7 12

LESSON CLOCK

25'

30'

30'

25'

10

Sharing an amount	25 min
From one share to the total	30 min
From a difference	30 min
Recipes and mixtures	25 min
Homework	10 min

LEARNING OBJECTIVES

- Share a quantity in a given ratio.
- Find the total from one share or from the difference of two shares.
- Scale a recipe or a mixture up and down.
- Check that the shares rebuild the total.

TEXTBOOK

National	pp. 121–130
Cambridge	Stage 7, pp. 118–125
Workbook	Exercise 12.3

01

Explanation

Sharing an amount

STEP		SHARING 120 IN 2 : 3	
add the parts		$2 + 3 = 5$	
one part		$120 \div 5 = 24$	
the shares		48 and 72	
check		$48 + 72 = 120 \checkmark$	

AMOUNT	RATIO	SHARES
120	2 : 3	48, 72
200	3 : 7	60, 140
450	4 : 5	200, 250
96	1 : 3	24, 72

THE CHECK TAKES TWO SECONDS

Adding the shares must give the original amount. Nearly every arithmetic slip in this topic shows up in that one line.

From one share to the total

PROBLEM	ONE PART	TOTAL
ratio 2 : 3, the first share is 18	9	45
ratio 3 : 4, the second share is 28	7	49
ratio 5 : 2, the second share is 16	8	56
ratio 1 : 6, the first share is 9	9	63

DIVIDE THE SHARE BY ITS OWN NUMBER, NOT BY THE TOTAL PARTS

With ratio 2 : 3 and a first share of 18, one part is $18 \div 2 = 9$ — not $18 \div 5$. The share belongs to the 2, so the 2 is the divisor.

From a difference

The difference between two shares is the difference of their parts, times one part.

PROBLEM	PARTS OF DIFFERENCE	ONE PART	TOTAL
ratio 5 : 3, difference 16	2	8	64
ratio 7 : 4, difference 12	3	4	44
ratio 3 : 1, difference 30	2	15	60

ONE PART ANSWERS EVERYTHING

Whether the problem gives a total, one share or a difference, the first move is the same: turn what you are given into the value of one part.

Recipes and mixtures

PROBLEM	WORKING	ANSWER
concrete $1 : 2 : 4$ cement, sand, gravel in 700 kg	one part = 100	100, 200, 400 kg
a recipe for 4 uses 600 g flour: for 6	$150 \cdot 6$	900 g
paint $3 : 2$ blue to white, 12 l of blue	one part = 4	8 l white
a drink $1 : 5$ in 1.8 l	one part = 0.3	0.3 l syrup

KEEP THE UNITS WITH EVERY NUMBER

A recipe scaled from kilograms to grams by mistake is out by a factor of a thousand. Writing the unit beside each share is the cheapest protection there is.

02

Worked examples

EXAMPLE 1 Share 450 in the ratio 4 : 5.

1

$4 + 5 = 9$ parts.

2

$450 \div 9 = 50$ for one part.

3

$4 \cdot 50 = 200$ and $5 \cdot 50 = 250$.
Check: $200 + 250 = 450$ ✓

ANSWER

200 and 250

EXAMPLE 2 Two shares are in the ratio $5 : 3$ and differ by 16. Find both.

① The difference is $5 - 3 = 2$ parts.

② $16 \div 2 = 8$ for one part.

③ $5 \cdot 8 = 40$ and $3 \cdot 8 = 24$.

Check: $40 - 24 = 16$ ✓

ANSWER

40 and 24

EXAMPLE 3 Concrete is mixed $1 : 2 : 4$ by mass. How much of each is in 700 kg?

① $1 + 2 + 4 = 7$ parts.

② $700 \div 7 = 100$ kg for one part.

③ 100, 200 and 400 kg.

Check: they total 700 ✓

ANSWER

100, 200, 400 kg

03 Interactive model

Use a real recipe card and scale it for the whole class; the ratio work is identical to the textbook and the purpose is never in doubt.

Total, share or difference?

120 in 2 : 3 gives:

40, 80

48, 72

50, 70

60, 60

200 in 3 : 7 gives:

60, 140

70, 130

80, 120

100, 100

Ratio 2 : 3 with first share 18: one part is:

3.6

6

9

18

So the total is:

27

45

30

90

Ratio 5 : 3 with difference 16: one part is:

2

8

16

4

So the shares are:

40 and 24

30 and 14

50 and 34

20 and 4

Concrete 1 : 2 : 4 in 700 kg has cement:

50 kg

100 kg

175 kg

200 kg

The check on any sharing problem is:

the ratio

the shares add to the total

the difference

the units

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
To share in a ratio	Nisbatda taqsimlash	Разделить в отношении
Share	Ulush	Доля
Difference	Ayirma	Разность
Recipe	Retsept	Рецепт
To scale	Miqyoslash	Масштабировать
Mixture	Aralashma	Смесь
Total	Jami	Всего
Check	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The first step in sharing is:

divide by the first part

add the parts

multiply

subtract

96 in $1 : 3$ gives:

24 and 72

32 and 64

48 and 48

16 and 80

Given one share, divide it by:

the total parts

its own number in the ratio

the other number

the total

Ratio 7 : 4 with difference 12: the total is:

A recipe for 4 scaled to 6 multiplies by:

Paint 3 : 2 with 12 l of blue needs white:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Share 120 in 2 : 3

ANSWER 48 and 72

2. Share 200 in 3 : 7

ANSWER 60 and 140

3. Share 450 in 4 : 5

ANSWER 200 and 250

4. Share 96 in 1 : 3

ANSWER 24 and 72

5. Ratio 2 : 3, first share 18: one part

ANSWER 9

6. And the total

ANSWER 45

7. Ratio 3 : 4, second share 28: the total

ANSWER 49

Medium · the standard question

7 PROBLEMS

1. Ratio 5 : 3, difference 16

ANSWER 40 and 24

2. Ratio 7 : 4, difference 12

ANSWER 28 and 16

3. Concrete 1 : 2 : 4 in 700 kg

ANSWER 100, 200, 400 kg

4. A recipe for 4 with 600 g flour: for 6

ANSWER 900 g

5. Paint 3 : 2, 12 l blue: the white

ANSWER 8 l

6. A drink 1 : 5 in 1.8 l: the syrup

ANSWER 0.3 l

7. Ratio 5 : 2, second share 16: the total

ANSWER 56

Hard · stretch and reason

7 PROBLEMS

1. Share 540 in 2 : 3 : 4

ANSWER 120, 180, 240

2. Ratio 4 : 9, difference 35: the total

ANSWER 91

3. A recipe for 6 uses 750 g: for 10

ANSWER 1250 g

4. Money shared 3 : 5 so that the smaller share is 45 000

ANSWER Total 120 000

5. Concrete 1 : 2 : 4 using 150 kg of cement: the total

ANSWER 1050 kg

6. Two numbers in ratio 4 : 7 add to 132

ANSWER 48 and 84

7. Why divide a known share by its own number?

ANSWER That share is exactly that many parts

07 Homework

Homework — 5 tasks

Add your shares back to the total before writing the answer down.

1. Share 360 in the ratio 5 : 4.
2. Share 840 in the ratio 1 : 2 : 4.
3. Two shares are in the ratio 7 : 3 and differ by 24. Find both.
4. A recipe for 5 people uses 400 g of rice. How much for 8?
5. Paint is mixed 4 : 3 blue to white. With 20 litres of blue, how much white is needed?